A VARIANCE FORM OF HAN'S BOUND ON THE FOURIER ENTROPY

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Abstract. Han proved that every Boolean function f satisfies

$$H(f) \le \frac{3}{\ln 2}\, I(f) + \sum_k \mathrm{Inf}_k(f) \log_2 \frac{4}{\mathrm{Inf}_k(f)},$$

where H is the Fourier entropy and I the total influence. We show that the logarithm may be taken relative to the variance:

$$H(f) \le \Big(2 + \frac{3}{\ln 2}\Big) I(f) + \sum_k \mathrm{Inf}_k(f) \log_2 \Big(1 + \frac{\mathrm{Var}(f)}{\mathrm{Inf}_k(f)}\Big).$$

The bound implies Han's up to an additive $I(f)$ and is smaller by a factor of order k for the AND of k variables. Written with $\log_2(1/\mathrm{Inf}_k(f))$ in the sum, Han's bound has the constant $2 + 3/\ln 2$ in front of $I(f)$; Li and Han lowered it to the sharp value $1/\ln 2$, and neither their bound nor ours implies the other. It follows that the Fourier entropy-influence conjecture holds, with constant $2 + 3/\ln 2 + \log_2(1 + 1/\varepsilon)$, for every function whose influences are all at least $\varepsilon \mathrm{Var}(f)$, and that $H(f) \le I(f)\,(3 + 3/\ln 2 + H(\pi))$, where $\pi_k = \mathrm{Inf}_k(f)/I(f)$ is the influence distribution. The proof combines the chain rule for the Fourier entropy along a decision tree with a one-coordinate estimate, and an adaptive version holds along any decision tree.

1. Introduction

Let $f : \{-1,1\}^n \to \{-1,1\}$ have Fourier expansion $f = \sum_S \hat{f}(S)\chi_S$. By Parseval $\mu(S) = \hat{f}(S)^2$ is a probability distribution on subsets of $[n]$, and we write

$$H(f) = \sum_S \mu(S) \log_2 \frac{1}{\mu(S)}, \qquad \mathrm{Inf}_k(f) = \sum_{S \ni k} \mu(S), \qquad I(f) = \sum_k \mathrm{Inf}_k(f),$$

and $\mathrm{Var}(f) = 1 - \hat{f}(\emptyset)^2$. The influence $\mathrm{Inf}_k(f)$ is also the probability that flipping x_k changes $f(x)$. The Fourier entropy-influence conjecture of Friedgut and Kalai [1] asserts that $H(f) \le C\, I(f)$ for an absolute constant C.

Several unconditional bounds of the form $H(f) \le I(f) \cdot$ (a logarithm) are known. Subadditivity of entropy gives $H(f) \le \sum_k h_2(\mathrm{Inf}_k(f))$, where h_2 is the binary entropy; Keller, Mossel and Schlank [7] deduce from it that $H(f) \le 2I(f)(1 + \log n - \log I(f))$. Kelman, Kindler, Lifshitz, Minzer and Safra [6] prove that the min-entropy is $O(I(f) \log I(f))$ for functions of constant variance, with a matching concentration of the spectrum. Chakraborty, Kulkarni, Lokam and Saurabh [3] and Wan, Wright and Wu [15] bound $H(f)$ in terms of decision-tree and communication complexity measures, and O'Donnell and Tan [10] show that the conjecture is preserved under suitable compositions; it holds for every linear threshold function [13]. Arunachalam, Chakraborty, Koucký, Saurabh and de Wolf [2] bound $H(f)$ by twice the average unambiguous parity-certificate complexity. Han [5, Theorem 1] proves $H(f) = O\big(I(f) + \sum_k \mathrm{Inf}_k(f) \log(1/\mathrm{Inf}_k(f))\big)$, and the last display of his proof gives the explicit form

$$H(f) \le \frac{3}{\ln 2}\, I(f) + \sum_k \mathrm{Inf}_k(f) \log_2 \frac{4}{\mathrm{Inf}_k(f)} = \Big(2 + \frac{3}{\ln 2}\Big) I(f) + \sum_k \mathrm{Inf}_k(f) \log_2 \frac{1}{\mathrm{Inf}_k(f)}. \tag{1}$$

Li and Han [8] proved, by an information-theoretic argument, that the constant $2+3/\ln 2$ in (1) may be replaced by the sharp value $1/\ln 2$, and that the resulting inequality holds for every real-valued f with $\sum_S \hat{f}(S)^2 = 1$. The last term of (1) is large when many influences are small. For functions of small variance this is forced, since every influence is at most $2\min(\Pr[f = 1], \Pr[f = -1]) \le \mathrm{Var}(f)$, and then the logarithm measures the size of $\mathrm{Var}(f)$ rather than how the influence is spread. Our main result replaces $1/\mathrm{Inf}_k$ in the logarithm by $\mathrm{Var}(f)/\mathrm{Inf}_k$.

Theorem 1. *For every $f : \{-1,1\}^n \to \{-1,1\}$,*

$$H(f) \le c_0\, I(f) + \sum_{k=1}^{n} \mathrm{Inf}_k(f) \log_2\Big(1 + \frac{\mathrm{Var}(f)}{\mathrm{Inf}_k(f)}\Big),$$

where $c_0 = 2 + 3/\ln 2 = 6.3281\ldots$ and a term with $\mathrm{Inf}_k(f) = 0$ is read as 0.

Since $\mathrm{Var}(f) \le 1$ and $\log_2(1+1/t) \le 1+\log_2(1/t)$ for $t \le 1$, Theorem 1 implies (1) with c_0+1 in place of c_0. Theorem 1 and the bound of Li and Han [8] do not imply each other: for majority their smaller constant makes theirs the smaller bound, while for AND_k (Example 8) the right side of theirs is $(k-1+1/\ln 2)\, I(f)$ and that of Theorem 1 is below $7.92\, I(f)$. Two consequences follow. The first is immediate from the theorem; the second also uses $\mathrm{Var}(f) \le I(f)$, the Poincaré inequality [9, Chapter 2].

Corollary 2. *If $\mathrm{Inf}_k(f) \ge \varepsilon \mathrm{Var}(f)$ for every relevant coordinate k, then $H(f) \le \big(c_0 + \log_2(1 + 1/\varepsilon)\big) I(f)$.*

Corollary 3. *Let $\pi_k = \mathrm{Inf}_k(f)/I(f)$ and $H(\pi) = \sum_k \pi_k \log_2(1/\pi_k)$. Then $H(f) \le \big(c_0 + 1 + H(\pi)\big) I(f)$.*

Proof. Since $\mathrm{Var}(f) \le I(f)$ and $\mathrm{Inf}_k(f) \le I(f)$,

$$\log_2\Big(1 + \frac{\mathrm{Var}(f)}{\mathrm{Inf}_k(f)}\Big) \le \log_2 \frac{2I(f)}{\mathrm{Inf}_k(f)} = 1 + \log_2 \frac{1}{\pi_k};$$

multiply by $\mathrm{Inf}_k(f)$ and sum. □

Corollary 3 has the shape of the bound of Keller, Mossel and Schlank, without their term $-2I(f)\log I(f)$, which helps when $I(f) > 1$ and hurts when $I(f) < 1$. Since $t\log_2(1+1/t) \le t + h_2(t)$, the right side of Theorem 1 never exceeds the subadditivity bound $\sum_k h_2(\mathrm{Inf}_k(f))$ by more than $(c_0+1)I(f)$, and it can be smaller by a factor of order $\log(1/\mathrm{Var}(f))$ when the variance is small (Example 8).

The proof is short. Fixing one coordinate splits the entropy into the average entropy of the two restrictions and a drop term D_k, and the drop has an exact decomposition into an erasure entropy [14] and a Jensen–Shannon divergence [12]. Lemma 5 bounds the drop by $\mathrm{Inf}_k\,(c_0 + \log_2(1 + \mathrm{Var}/\mathrm{Inf}_k))$ for every Boolean function. Summing along a decision tree that queries the coordinates in a fixed order expresses $H(f)$ as an average of such drops over restrictions, and the bound follows from the concavity of $(t, v) \mapsto t\log_2(1 + v/t)$. Querying the coordinates adaptively gives a stronger statement (Theorem 6). Section 4 works out examples and records numerical checks.

2. The drop of one coordinate

Fix a coordinate i and write $f_\pm = f_{i\to\pm1}$ for the two restrictions, functions of the other $n-1$ variables. Put

$$g = \tfrac{1}{2}(f_+ + f_-), \qquad h = \tfrac{1}{2}(f_+ - f_-),$$

so that $f = g + x_i h$, $\hat{f}(T) = \hat{g}(T)$ and $\hat{f}(T \cup \{i\}) = \hat{h}(T)$ for $T \not\ni i$. Since f is Boolean, $h(x) \in \{-1, 0, 1\}$ and $h(x) \neq 0$ exactly when i is pivotal at x, so

$$\sum_T \hat{h}(T)^2 = \mathbb{E}h^2 = \mathrm{Inf}_i(f), \qquad |\hat{h}(\emptyset)| \le \mathbb{E}|h| = \mathrm{Inf}_i(f). \tag{2}$$

Also g is the average of f over x_i, so $\mathrm{Var}(g) \le \mathrm{Var}(f)$, that is, $\sum_{T \neq \emptyset} \hat{g}(T)^2 \le \mathrm{Var}(f)$.

Let U have law μ. The drop of i is

$$D_i(f) = H(f) - \tfrac{1}{2}H(f_+) - \tfrac{1}{2}H(f_-).$$

The chain rule $H(U) = H(U \setminus \{i\}) + H(\mathbf{1}[i \in U] \mid U \setminus \{i\})$, together with the fact that the law of $U \setminus \{i\}$ is the equal mixture of the spectral distributions of f_+ and f_- (as $\hat{f}_\pm(T) = \hat{g}(T) \pm \hat{h}(T)$, and $\frac{1}{2}(\hat{g} + \hat{h})^2 + \frac{1}{2}(\hat{g} - \hat{h})^2 = \hat{g}^2 + \hat{h}^2$), gives

$$D_i(f) = E_i(f) + J_i(f),$$

where $E_i(f) = H(\mathbf{1}[i \in U] \mid U \setminus \{i\})$ and $J_i(f)$ is the Jensen–Shannon divergence between the spectral distributions of f_+ and f_-. In coordinates,

$$E_i(f) = \sum_{T \not\ni i} \Phi\big(\hat{g}(T)^2, \hat{h}(T)^2\big), \qquad \Phi(p, q) = p \log_2 \frac{p+q}{p} + q \log_2 \frac{p+q}{q},$$

with $\Phi(p, 0) = \Phi(0, q) = 0$. The function Φ is $(p+q)\, h_2(q/(p+q))$; it is nonnegative, positively homogeneous of degree one, concave (it is the perspective of the concave function h_2; directly, its Hessian is $-\frac{1}{(p+q)\ln 2}\left(\begin{smallmatrix} q/p & -1 \\ -1 & p/q \end{smallmatrix}\right) \preceq 0$), and nondecreasing in each argument. A concave function that is homogeneous of degree one is superadditive: $\Phi(p+p', q+q') = 2\Phi(\frac{p+p'}{2}, \frac{q+q'}{2}) \ge \Phi(p, q) + \Phi(p', q')$.

Lemma 4 ([12, Lemma 3])**.** *$J_i(f) \le \frac{2}{\ln 2}\mathrm{Inf}_i(f)$.*

Lemma 5. *For every Boolean f and every coordinate i, with $\delta = \mathrm{Inf}_i(f)$,*

$$E_i(f) \le \Big(2 + \frac{1}{\ln 2}\Big)\delta + \delta \log_2\Big(1 + \frac{\mathrm{Var}(f)}{\delta}\Big), \qquad D_i(f) \le c_0\, \delta + \delta \log_2\Big(1 + \frac{\mathrm{Var}(f)}{\delta}\Big).$$

Proof. Separate the term $T = \emptyset$ and apply superadditivity to the rest:

$$E_i(f) \le \Phi\big(\hat{g}(\emptyset)^2, \hat{h}(\emptyset)^2\big) + \Phi(A, B), \qquad A = \sum_{T \neq \emptyset} \hat{g}(T)^2 \le \mathrm{Var}(f), \quad B = \sum_{T \neq \emptyset} \hat{h}(T)^2 \le \delta,$$

using (2). For the first term, $h_2(x) \le 2\sqrt{x(1-x)}$ gives $\Phi(p, q) \le 2\sqrt{pq}$, so it is at most $2|\hat{g}(\emptyset)||\hat{h}(\emptyset)| \le 2\delta$ by (2) and $|\hat{g}(\emptyset)| \le 1$. For the second, monotonicity gives $\Phi(A, B) \le \Phi(\mathrm{Var}(f), \delta)$ and

$$\Phi(V, \delta) = V \log_2\Big(1 + \frac{\delta}{V}\Big) + \delta \log_2\Big(1 + \frac{V}{\delta}\Big) \le \frac{\delta}{\ln 2} + \delta \log_2\Big(1 + \frac{V}{\delta}\Big),$$

since $\log_2(1 + y) \le y/\ln 2$. This proves the bound on E_i; the bound on D_i adds Lemma 4, and $2 + 1/\ln 2 + 2/\ln 2 = c_0$. □

The proof uses only one distinction among the sets T, whether T is empty. Up to the term $(2+1/\ln 2)\delta$ it sharpens the subadditivity estimate $E_i(f) \le h_2(\mathrm{Inf}_i(f)) \approx \mathrm{Inf}_i(f) \log_2(1/\mathrm{Inf}_i(f))$ by replacing $1/\mathrm{Inf}_i$ with $\mathrm{Var}/\mathrm{Inf}_i$ inside the logarithm.

3. Summing along a decision tree

A decision tree that queries coordinates of f until the restricted function is constant defines, for a uniformly random input, a random sequence of restrictions $f = f_0, f_1, \dots, f_m$, where f_j is f restricted by the first j answers and f_m is constant. Let i_j be the coordinate queried at step j. Since H of a constant function is 0 and the next answer is uniform given the past, the definition of the drop gives, by induction on the depth,

$$H(f) = \mathbb{E}\sum_{j\ge 1} D_{i_j}(f_{j-1}), \qquad I(f) = \mathbb{E}\sum_{j\ge 1} \mathrm{Inf}_{i_j}(f_{j-1}), \tag{3}$$

the second because $\mathrm{Inf}_k(f) = \frac{1}{2}\mathrm{Inf}_k(f_+) + \frac{1}{2}\mathrm{Inf}_k(f_-)$ for $k \neq i$ and $I(f) = \mathrm{Inf}_i(f) + \frac{1}{2}I(f_+) + \frac{1}{2}I(f_-)$ (compare the splitting lemma for influences in [4, Section 2]).

Theorem 6. *For every Boolean f and every decision tree as above,*

$$H(f) \le c_0\, I(f) + \mathbb{E}\sum_{j\ge 1} \mathrm{Inf}_{i_j}(f_{j-1}) \log_2\Big(1 + \frac{\mathrm{Var}(f_{j-1})}{\mathrm{Inf}_{i_j}(f_{j-1})}\Big).$$

Proof. Apply Lemma 5 to each term of (3) and use the second identity in (3). □

Proof of Theorem 1. Use the tree that queries $1, 2, \dots, n$ in this order, skipping nothing; once f_{j-1} is constant the remaining terms vanish. Then f_{k-1} is f with $x_1, \dots, x_{k-1}$ fixed uniformly at random, and

$$\mathbb{E}\,\mathrm{Inf}_k(f_{k-1}) = \mathrm{Inf}_k(f), \qquad \mathbb{E}\,\mathrm{Var}(f_{k-1}) = \mathrm{Var}(f) - \mathrm{Var}\big(\mathbb{E}[f \mid x_1, \dots, x_{k-1}]\big) \le \mathrm{Var}(f).$$

The function $\psi(t, v) = t\log_2(1 + v/t)$, with $\psi(0, v) = 0$, is the perspective of the concave function $v \mapsto \log_2(1 + v)$, hence jointly concave on $[0, \infty)^2$, and it is nondecreasing in v. By Jensen's inequality

$$\mathbb{E}\,\psi\big(\mathrm{Inf}_k(f_{k-1}), \mathrm{Var}(f_{k-1})\big) \le \psi\big(\mathrm{Inf}_k(f), \mathbb{E}\mathrm{Var}(f_{k-1})\big) \le \psi\big(\mathrm{Inf}_k(f), \mathrm{Var}(f)\big).$$

Summing over k and using Theorem 6 proves the theorem. □

Remark 7. The same argument with $\mathrm{Var}(f_{j-1})$ replaced by 1, using the concavity of $t \mapsto t\log_2(1 + 1/t)$ in place of that of ψ, gives Han's bound (1) up to one unit in the constant. Han's proof runs through the moments $\sum_S |\hat{f}(S)|^{2+2\epsilon}$ of restricted functions, revealed one coordinate at a time, and differentiates at $\epsilon = 0$; the route above works with the entropy directly.

4. Examples and checks

Example 8. Let $f = \mathrm{AND}_k$ take the value -1 only at the all-ones input. Then $\hat{f}(\emptyset) = 1 - 2^{1-k}$ and $\hat{f}(S) = -2^{1-k}$ for $S \neq \emptyset$, so $\mathrm{Inf}_i(f) = 2^{1-k}$, $I(f) = k2^{1-k}$ and $\mathrm{Var}(f) = 2^{2-k}(1 - 2^{-k})$. The ratio $\mathrm{Var}(f)/\mathrm{Inf}_i(f)$ is below 2, and Theorem 1 gives $H(f) \le (c_0 + \log_2 3)\, I(f) < 7.92\, I(f)$, while the right side of (1) is $(k + 1 + 3/\ln 2)\, I(f)$. The true value is $H(f)/I(f) = 4(k-1)/k + 2/(k\ln 2) + o(1/k)$, which tends to 4. The same holds for OR_k.

Example 9. For majority on n variables, $\mathrm{Var} = 1$ and every influence is of order $n^{-1/2}$, so Theorem 1 and Han's bound both give order $\sqrt{n}\log n$, while $H = O(\sqrt{n})$ [11]; the same happens for balanced tribes. For these functions the variance is bounded below and the loss is in the logarithm itself. A heuristic computation along the tree that queries the coordinates of majority in order suggests that the adaptive form does not remove it. A proof of the conjecture along these lines needs a step estimate $D_i \le C\,\mathrm{Inf}_i$ at some coordinate of each restriction, which is the one-coordinate form of the conjecture [4, 12].

Numerical checks. On 1720 random Boolean functions on 2 to 10 variables, with values -1 taken independently with probability 0.5, 0.2, 0.05 or 0.01, the ratio of $H(f)$ to the right side of Theorem 1 is at most 0.486 on each of these functions. Over all nonconstant monotone Boolean functions of six variables the maximum of this ratio is 0.4735, and over those of seven variables it is 0.4812, attained by OR_n and AND_n; the maxima of the ratio of $H(f)$ to the right side of (1) are 0.3870 and 0.3935. Over 4,767 coordinate steps of random functions on 2 to 8 variables, with D_i, E_i and J_i computed separately, the identity $D_i = E_i + J_i$ holds to within $3 \cdot 10^{-15}$, and the three bounds of Lemmas 4 and 5 are attained to within the factors 0.869, 0.599 and 0.677; along 928 random adaptive trees the ratio of $H(f)$ to the right side of Theorem 6 is at most 0.484. The scripts are in the repository listed below.

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Declaration of competing interest

The author holds equity in Heights Labs and in Convex Nexus Capital and has a management role at Convex Nexus Capital. These entities are included in the funding declaration above.

Declaration of generative AI and AI-assisted technologies

The paper comes out of Math Brain, an ongoing project on the use of AI systems for research mathematics, and its results were developed with the Brain [18] and Math Brain [16]. In November 2026 a Claude Fable agent (Anthropic), connected to the Brain and working alongside Codex agents running OpenAI's GPT-6.1 on the same problem, derived the decomposition of the drop, the path formula and the one-coordinate estimate of Lemma 5; a Codex agent refereed the estimate and found the shorter bound for the empty set. Claude Opus 5.5 (Anthropic) searched the literature, proved Theorems 1 and 6 and the corollaries, wrote the random-function check, prepared the manuscript and audited it; a second Claude Opus 5.5 agent ran the exhaustive computation over monotone functions on six and seven variables. These checks are not peer review. The author directed the work and is responsible for the content of the paper and for decisions concerning its dissemination.

Code and data availability

Code and outputs: `https://github.com/yspennstate/fei-variance-han-bound`.

The paper is part of the Math Brain project, whose public repository is `https://github.com/yspennstate/math-brain-public` [16].

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